

Muhammad Farhan

AI Engineer — LLMs & Agentic Systems · Generative AI · Python · FastAPI

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PROFESSIONAL SUMMARY

Computer Science graduate (all coursework completed, degree pending conferral) and self-driven AI Engineer specialising in LLMs and Agentic AI systems, with 1+ year of hands-on experience designing and deploying multi-agent architectures using LangChain and LangGraph. Experienced in orchestrating collaborative agent workflows — from autonomous content-generation pipelines to real-time conversational agents that evaluate and respond to human input — and in serving them behind production-style FastAPI endpoints. Grounded in strong classical ML and deep learning foundations (CNNs, predictive modelling, reinforcement learning), giving depth beneath the agentic work rather than prompt-level familiarity alone. Focused on building AI systems that solve real user pain points and translate into measurable impact.

TECHNICAL SKILLS

AI / ML	Predictive Modelling · Classification & Regression · Feature Engineering · Model Evaluation (Accuracy, Precision, Recall, F1, ROC-AUC) · Scikit-learn · Pandas · NumPy · Matplotlib
Deep Learning	CNNs · NLP · Perceptron / Neural Networks · Reinforcement Learning · TensorFlow / Keras
Generative AI	LLM Integration · Multi-Agent Systems · Prompt Engineering · LangChain · LangGraph · RAG (foundations)
Backend & APIs	FastAPI · REST API Design · Streamlit · Flask
Web Development	MERN Stack (MongoDB · Express.js · React · Node.js) · JavaScript · Next.js · REST Integration
Tools & Infra	Python (Advanced) · Git & GitHub · Docker · Jupyter · Redis
Familiar	PostgreSQL · n8n workflow automation
Languages	English (Professional) · Urdu (Native) · Pashto (Native)

FEATURED PROJECTS

Multi-Agent Blog Generation System — Final Year Project (Team of 2) 2025 – 2026

Stack: Python · LangGraph / LangChain · LLMs · FastAPI · Next.js

- Designed and built a multi-agent Generative AI system in which specialised agents collaborate to research, outline, draft, and refine long-form blog content end to end.
- Implemented agent orchestration and state handoff so that each agent consumes the previous agent's output, producing coherent, structured articles from a single topic prompt.
- Exposed the pipeline through a FastAPI backend with a web interface, enabling non-technical users to generate publish-ready content on demand.

AI Court Simulation & Debate Evaluation System — Capstone Project, GIKI 2026 – Present

Stack: Python · LLMs · Multi-Agent Architecture · FastAPI

- Building an AI-powered moot court platform where law students argue cases in real time against an opposing-counsel LLM agent, creating repeatable courtroom practice without a human partner.
- Developing an autonomous AI judge agent that scores argument quality, legal reasoning, and rebuttal strength, returning structured feedback so students can measure improvement across sessions.
- Architecting the multi-agent conversation flow and evaluation rubric to keep debate turns contextual, on-topic, and consistently graded.

Kidney Disease Classification — Deep Learning (Computer Vision) 2025

Stack: Python · TensorFlow / Keras · CNN · NumPy · Matplotlib

- Trained a Convolutional Neural Network to classify kidney disease from medical scan images, covering the full pipeline from image preprocessing and augmentation to training and evaluation.
- Tuned model architecture and hyperparameters, benchmarking performance with accuracy, precision, recall, and confusion-matrix analysis to reduce false negatives on clinical-style data.

Reinforcement Learning Stick-Fighter Game — Personal Project 2025

Stack: Python · Reinforcement Learning · NumPy

- Built a fighting game featuring an RL agent that learns and adapts its strategy through repeated combat against a human player, improving over successive matches.

- Designed the reward function, action space, and training loop, then evaluated agent progression by tracking win-rate improvement over episodes.

Customer Churn Prediction — *Machine Learning Project* 2025

Stack: Python · Scikit-learn · Pandas · FastAPI

- Developed a model predicting customer churn from behavioural, transactional, and demographic signals, enabling proactive retention targeting.
- Executed data preprocessing, feature engineering, and comparative model evaluation using accuracy, precision, recall, and ROC-AUC.

ADDITIONAL MACHINE LEARNING PROJECTS

- URL Phishing Detector — Built a classification model that identifies malicious URLs from lexical and structural features, applying feature engineering to separate phishing from legitimate links.
- House Price Prediction — Regression model for property price estimation with outlier handling, feature scaling, and RMSE/R² based evaluation.

EDUCATION

BS Computer Science · *University of Peshawar, Pakistan* 2022 – 2026

- All 8 semesters completed; degree awaiting official conferral. Available for full-time roles immediately.
- Relevant coursework: Machine Learning · Deep Learning · Data Structures & Algorithms · Databases · Software Engineering
- Final Year Project: Multi-Agent Blog Generation System using LLM agent orchestration.

CERTIFICATIONS & TRAINING

- Fundamentals of Deep Learning — NVIDIA (Coursera)
- Fundamentals of Machine Learning — NVIDIA (Coursera)
- AI / Agentic AI Capstone Program — GIKI (Ghulam Ishaq Khan Institute) · 2026
- MERN Stack Development — NAVTTC (National Vocational & Technical Training Commission)
- Web and App Development — Saylani Mass IT Training (SMIT)

CAREER FOCUS

Actively pursuing AI/ML Engineer roles with a focus on Generative AI and Agentic AI systems. Planning postgraduate specialisation in GenAI and Agentic AI.